

Mini Project Report on
QuizGuard Using API

Submitted in partial fulfillment of the requirements for the degree of
BACHELOR OF ENGINEERING
IN

Computer Science & Engineering
Artificial Intelligence & Machine Learning By

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Parshvanath Charitable Trust's
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CERTIFICATE

This is to certify that the project entitled “**Quiz Guard Using API**” is a bonafide work of **Soham Jathar** (23106084), **Onkar Deshmukh** (23106006), **Bhavika Kadam** (23106079), **Prathamesh Kolhe** (23106132) submitted to the University of Mumbai in partial fulfillment of the requirement for the award of **Bachelor of Engineering in Computer Science & Engineering (Artificial Intelligence & Machine Learning)**.

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Project Report Approval

This Mini project report entitled “QuizGuard Using API” by **Soham Jathar, Onkar Deshmukh, Bhavika Kadam and Prathamesh Kolhe** is approved for the degree of *Bachelor of Engineering in Computer Science & Engineering, (AIML) 2024-25*.

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Date:

Declaration

We declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

The API-powered quiz exam system is a cutting-edge assessment solution that leverages artificial intelligence (AI) to provide a secure, efficient, and personalized experience for students. At the heart of the system is a sophisticated API engine that analyzes student responses and provides instant feedback. This feature enables students to receive timely and accurate feedback on their performance, allowing them to identify areas of improvement and adjust their learning strategies accordingly.

One of the most significant advantages of the API-powered quiz exam system is its ability to ensure academic integrity. The system incorporates advanced face detection technology to verify the identity of students taking the exam. This feature prevents impersonation and ensures that the student taking the exam is the same person who registered for the course. Additionally, the system alerts the instructor or proctor if a student attempts to switch tabs or leave the exam window, preventing academic dishonesty and ensuring that students remain focused on the exam.

The API-powered quiz exam system also provides real-time monitoring capabilities, allowing instructors or proctors to track student activity during the exam. This feature enables prompt intervention if suspicious behavior is detected, ensuring that the assessment process remains fair and secure. Furthermore, the system provides automated grading and feedback, reducing the workload for instructors and enabling them to focus on teaching and mentoring.

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CHAPTER 1

INTRODUCTION

1. INTRODUCTION

A **secure online quiz-taking app** is a powerful platform that enables users to take quizzes or exams remotely while maintaining the highest standards of **integrity, privacy, and security**. These apps have become essential tools in education, corporate training, and certification programs, ensuring a seamless and controlled assessment experience. To uphold credibility, they incorporate **robust security measures**, such as user authentication, real-time monitoring, and anti-cheating mechanisms, which prevent dishonest practices and maintain fairness. Features like **time limits, randomized questions, and encrypted data storage** further enhance security, protecting both the participants and the quiz content. By leveraging modern technology, these platforms offer a user-friendly interface for learners, educators, and organizations, making it easier to assess knowledge and skills in a trustworthy environment.

The **integration of API in education** has revolutionized the way assessments are conducted, making online quizzes more **efficient, personalized, and data-driven**. API -powered quiz systems utilize advanced algorithms to generate **adaptive assessments** that adjust based on an individual student's learning pace and ability. This ensures that each test is appropriately challenging, catering to different skill levels and optimizing the learning experience. API -driven platforms are particularly beneficial in academic settings, professional development programs, and certification exams, where personalized learning paths can significantly enhance knowledge retention and understanding.

Beyond just test customization, API also **automates grading**, providing **instant feedback** to students. This immediate response is crucial for reinforcing learning, as it helps learners understand their mistakes and improve in real time. Additionally, API tools can **analyze vast amounts of student performance data**, offering educators deeper insights into **trends, misconceptions, and areas requiring additional support**. These intelligent analytics help teachers and trainers identify common challenges among students, allowing for targeted interventions and personalized recommendations.

Moreover, API -powered quiz platforms support a variety of **question formats**, including multiple-choice, true/false, short-answer, and even **API -assisted open-ended evaluations**, making assessments more comprehensive. Some advanced systems even use **natural language processing (NLP)** to evaluate written responses, further reducing the need for manual grading.

By integrating API , online quiz-taking is no longer just a tool for measuring knowledge—it has evolved into an essential part of the learning journey. It **streamlines administrative tasks**, enhances **engagement**, and promotes a **more dynamic and interactive educational experience**. As API continues to advance, the future of online assessments will become even more intelligent, **making learning more accessible, efficient, and impactful for all**.

CHAPTER 2
LITERATURE SURVEY

2. LITERATURE SURVEY

2.1 HISTORY

The adoption of online proctoring technology for remote examinations represents a significant shift from traditional exam-taking methods. Historically, examinations were conducted in physical classrooms under direct human supervision, ensuring integrity but limiting accessibility. Manual invigilation required extensive logistical planning, increasing costs and administrative burdens.

With the rise of digital education and remote learning, the need for secure online assessments became evident. Early solutions relied on webcam monitoring, but concerns over cheating and impersonation led to the integration of **AI-driven proctoring, biometric verification, and real-time screen tracking**. The development of **WebRTC** for real-time video streaming and advancements in **machine learning** have enhanced automated exam monitoring, making online proctoring more reliable and scalable.

2.2 LITERATURE REVIEW

1) API-powered Online Exam Proctoring Systems (IEEE EXPLORE 2021)

S. Kumar, R. Gupta, A. Sharma

This paper discusses the development of an API-powered online exam proctoring system designed to enhance the security and integrity of remote assessments. The system utilizes machine learning techniques to analyze student behavior, detect suspicious activities, and prevent cheating. The authors highlight the effectiveness of facial recognition, gaze tracking, and keystroke dynamics in reducing academic dishonesty while also addressing challenges such as false positives and privacy concerns.

2) Automated Remote Exam Monitoring: Challenges and Solutions (SpringerLink 2020)

M. Al-Baity, N. Hossain

The study explores the effectiveness of automated proctoring solutions in online examinations. It discusses the role of API-based monitoring tools in detecting cheating behaviors, such as looking away from the screen or using unauthorized resources. The paper also examines the limitations of current systems, including issues related to accessibility, algorithmic bias, and student privacy. The authors suggest incorporating human oversight alongside API to improve accuracy and fairness.

3) API-based Proctoring for Secure Online Assessments (IEEE EXPLORE 2019)

J. Lee, H. Kim, Y. Park

This research focuses on the use of AI-driven proctoring tools in ensuring the security of online examinations. The study outlines how machine learning algorithms analyze students' facial expressions, voice activity, and typing patterns to detect anomalies. The paper highlights the system's ability to minimize human intervention, increase efficiency, and provide real-time alerts to instructors. The authors also discuss the ethical concerns surrounding API-based surveillance and the need for more transparent policies.

4) A Case Study on the Implementation of API Proctoring in Universities (Journal of Computer Applications 2018)

A. Singh, P. Patel

This case study examines the deployment of an API-powered proctoring system in a university setting. The paper provides insights into student experiences, technical

challenges, and administrative benefits. The study highlights the system's ability to reduce manual proctoring efforts while improving exam integrity. However, it also notes student discomfort with constant API surveillance and suggests future improvements, such as integrating explainable API to enhance trust.

5) Digital Exam Proctoring: A Comparative Analysis of Methods (International Journal of Digital Systems 2021)

R. O'Connor, M. Johnson

This paper reviews various digital proctoring approaches, including API-based and human-involved monitoring. It evaluates their effectiveness in preventing academic misconduct and discusses the trade-offs between security and student privacy. The study provides a comparative analysis of different proctoring technologies, such as face recognition, eye movement tracking, and plagiarism detection, and offers recommendations for optimizing online assessment systems.

6) The Role of API in Modern Online Exam Proctoring (IEEE EXPLORE 2020)

L. Brown, T. Green

This study examines the role of API in online exam proctoring, focusing on advancements in deep learning and natural language processing for fraud detection. The authors explore how API models analyze speech patterns and detect whispered conversations during online exams. The research also discusses the potential for future proctoring systems to incorporate blockchain technology for securing exam records and preventing result tampering.

CHAPTER 3

Problem Statement

3. Problem Statement

Traditional methods of conducting online exams face significant challenges in maintaining academic integrity, ensuring fairness, and providing a seamless experience for both students and educators. Conventional proctoring methods, such as in-person invigilation or basic online monitoring, are often inefficient, prone to human error, and incapable of effectively preventing cheating in remote assessments. These methods also struggle with scalability, making it difficult for institutions to manage large-scale exams without excessive administrative overhead.

Furthermore, existing online exam systems lack intelligent mechanisms to detect suspicious behavior in real-time. Simple webcam monitoring or time-limited assessments do not provide robust security, leaving gaps that tech-savvy students can exploit. Issues such as impersonation, unauthorized resource usage, and screen switching remain prevalent in remote exams. Additionally, students often face privacy concerns when subjected to intrusive surveillance methods, which can lead to discomfort and hinder their performance.

The core problem, therefore, lies in the absence of a **smart, AI-driven proctoring system** that can intelligently monitor exams, detect potential misconduct, and ensure a fair testing environment without compromising student privacy. There is also a gap in leveraging API-powered analytics to provide deeper insights into student behavior, exam performance, and potential patterns of malpractice.

To address these challenges, the need arises for a **secure, adaptive, and API-integrated online exam proctoring system** that can:

- **Automate invigilation** Using API-based face and behavior recognition.
- **Detect anomalies** such as eye movement, multiple faces, or background noise.
- **Prevent unauthorized activities** like screen switching or external assistance.
- **Ensure a balance between security and privacy** by offering non-intrusive monitoring techniques.
- **Provide real-time alerts and post-exam analytics** for educators to assess exam integrity effectively.

By implementing such a system, institutions can enhance the credibility of online examinations, reduce administrative workload, and create a secure, fair, and seamless assessment process for all stakeholders.

CHAPTER 4

Experimental Setup

4. Experimental Setup

4.1 Hardware Setup

To effectively implement an API-powered online exam proctoring system, the following hardware components are essential for ensuring secure, efficient, and scalable remote assessments:

1. Candidate Devices (Laptops, Desktops, or Tablets)

- **Type:** Personal computing devices used by students to take exams.
- **Minimum Configuration:**
 - **Processor:** Intel Core i5 (or equivalent) for smooth performance.
 - **RAM:** Minimum 8 GB for handling live monitoring and exam software.
 - **Storage:** At least 256 GB SSD to support exam files and real-time processing.
 - **Camera:** Minimum 720p HD webcam for identity verification and monitoring.
 - **Microphone:** Built-in or external mic for audio monitoring.
 - **Operating System:** Windows 10 or later, macOS, or compatible Linux distributions.
- **Usage:** Candidates will use these devices to access the exam portal, authenticate themselves, and complete the test under AI-monitored conditions.

2. Proctoring Servers

- **Type:** Dedicated or cloud-based servers responsible for handling exam sessions and AI-driven monitoring.
- **Configuration:**
 - **Processor:** Multi-core processor (Intel Xeon or equivalent) for processing multiple live video streams.
 - **RAM:** Minimum 32 GB to handle concurrent student sessions efficiently.
 - **Storage:** 1 TB SSD or more for storing video logs, exam responses, and analytics data.
 - **Operating System:** Linux-based server OS (Ubuntu Server, CentOS) or Windows Server.
- **Usage:** These servers process and analyze live video feeds, log exam activity, and detect anomalies Using API algorithms.

3. Network Infrastructure

- **Type:** Stable wired or wireless internet connection for seamless exam conduction.
- **Components:**
 - **Router/Switch:** High-speed routers to manage exam traffic and real-time proctoring data exchange.
 - **Bandwidth Requirements:** Minimum **5 Mbps per candidate** for smooth video streaming and low-latency responses.
 - **Firewall & Security Measures:** Ensure encrypted connections and prevent unauthorized access.
- **Usage:** Facilitates secure communication between students' devices, proctoring servers, and examiners.

4. API Processing & Cloud Services (If Cloud-Based)

- **Type:** Cloud computing platform for handling AI-driven exam monitoring and fraud detection.
- **Recommended Services:** AWS, Google Cloud, or Microsoft Azure.
- **Configuration:**
 - **GPU-Enabled Instances:** Required for deep learning models that analyze facial recognition and behavior detection.
 - **Data Storage:** Secure cloud storage (minimum 5 TB) for exam recordings and analytics.
 - **Encryption:** AES-256 encryption for securing exam data and student identities.
- **Usage:** API-based analysis of exam behavior, suspicious activities, and real-time alerts for potential misconduct.

4.2 Software Setup

1. API-powered Proctoring & Exam Monitoring

Facial Recognition & Identity Verification

- **Libraries/Tools:** OpenCV.js (for face tracking in the browser).
- **Usage:** Ensures only the registered student is taking the exam.

Cheating Detection & Behavioral Analysis

- **AI Models:** TensorFlow.js (for gaze tracking and movement detection).
- **Real-Time Monitoring:** OpenCV.js (for eye tracking), Medi API pe (for face detection).
- **Usage:** Detects suspicious activities like multiple faces, unauthorized materials, or absence from the screen.

Screen Activity & Browser Restriction

- **Libraries:** JavaScript API s (Browser Visibility API , Page Focus API).
- **Usage:** Prevents tab switching, detects unusual behavior, and blocks unauthorized activities within the exam environment.

2. Backend Server

- **Framework:** JavaScript (No backend server required as everything runs on the browser).
- **Database:** Browser Local Storage (LocalStorage, IndexedDB for structured exam data).
- **Usage:** Stores exam details, student records, and logs locally in the browser.

3. Frontend Interface

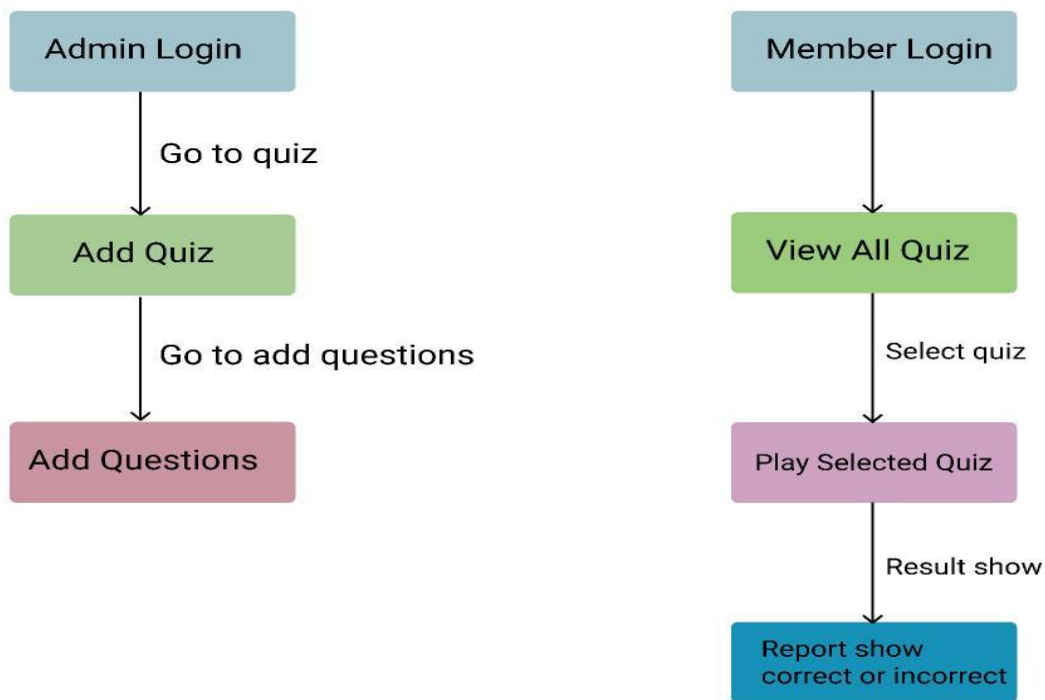
- **Languages Used:** HTML, CSS, JavaScript.
- **Frameworks/Libraries:**
 - Vanilla JavaScript for dynamic UI.
 - WebRTC for live webcam monitoring.
 - IndexedDB API for storing exam logs.

CHAPTER 5

Proposed System & Implementation

5. Proposed system & Implementation

5.1 Block Diagram of Proposed System



5.2 Description of Block Diagram

Admin Panel

- Authorized actions for the admin include:
 - Creating and managing quizzes.
 - Setting time limits and restrictions.
 - Viewing user activity logs.
- These actions interact with the local storage database for data management.

Student Panel

- Authorized actions for students include:
 - Registering for quizzes.
 - Taking quizzes within the assigned time limit.
 - Receiving quiz scores and feedback.
- This panel ensures smooth user interaction and secures exam integrity.

5.1 Implementation

Brand Logo
Institute Code

Institute ID

Password

[Student Login Form](#)
Sign In

[Home](#) [Contact](#) **Sign up**



[Home](#)

Logout

Welcome Mr : A P Shah Institute of tech

Teachers

200

Students

2000

Subjetc

8

Create Subject

Create Subject ^

Show Subject

Update And Delete All Subject ^

Create Subject Related Questions

Create Question ^

Update & Delete Subject Related Questions

Update & Delete Question ^

Students & Teachers Registration Form

Registration form ^

Students And Teacher Regarding Data

Welcome to online quiz examination center



Student's & Teacher's Login

Choose Space Code

Choose Space Code

Institute ID

Password

Login

Admin Login

Follow The Given Guideline

Avoid Tab switching

Sit at the silent place

Don't use any cheating device

NO COPY !!

BE HAPPY !!

C Programming

Start Quiz

Q-1 : What is c programming

- ☐ computer language
- ☐ animal
- ☐ bird
- ☐ fruit

Next

5.2 Advantages

1) Improved Accuracy and Reduced Errors

- **Automated Quiz Management:** Eliminates manual quiz handling, reducing the chances of human error in grading and tracking.
- **Secure Data Handling:** Local storage ensures data integrity by preventing unauthorized modifications or loss.

2) Enhanced Efficiency

- **Time-Restricted Quizzes:** Automated time limits ensure fairness and prevent extended access to quiz sessions.
- **Seamless Quiz Administration:** Admins can create and manage quizzes efficiently, reducing administrative workload.

3) User-Friendly

- **Simple Quiz Participation:** Students can easily register, take quizzes, and receive feedback without complex setups.
- **Instant Results and Feedback:** Users get immediate scores and feedback, enhancing the learning experience.

CHAPTER 6

Conclusion

6. Conclusion

Conclusion

The implementation of QuizGuard has demonstrated a reliable and efficient solution for conducting online quizzes while ensuring exam integrity. By integrating features such as time-restricted access, secure local storage, and automated result generation, the system streamlines the quiz-taking process for both students and administrators. The user-friendly interface enables students to participate seamlessly, while admins can efficiently create and manage quizzes without excessive manual effort.

With its ability to minimize errors, prevent cheating through access restrictions, and provide instant feedback, QuizGuard enhances the overall learning experience. The system's lightweight design and ease of use make it a practical solution for educational institutions seeking a secure and automated approach to online assessments.

Future Scope

While QuizGuard offers significant advantages, several potential enhancements could be implemented in the future. Future developments may include API-powered cheating detection, biometric authentication for increased security, and deeper analytics to assess student performance trends. Additionally, expanding compatibility with cloud storage solutions could further enhance accessibility and data security.

By continuously adapting to technological advancements and evolving educational needs, QuizGuard has the potential to become a more sophisticated and robust online examination platform.

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